Trojan Detection in Neural Networks

Neural Cleanse & DeepInspect–Neural Cleanse Evaluation on CIFAR-10

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Backdoor (Trojan) Attacks in Neural Networks

How the Attack Works:

- Model performs normally on **clean inputs**
- A small **trigger** injected into the image causes **targeted misclassification**
- Attack is embedded during training

Normal Behaviour

Label: **Stop**

Poisoned Behaviour

Label: **Go**

Why it's Dangerous:

- Appears **trustworthy** on during normal testing
- Trigger is small and hard to detect
- Only activates when attacker wants

Our Objective

Evaluate how effectively Neural Cleanse and DeepInspect-Neural Cleanse can detect hidden backdoor attacks

Experimental Setup and Attack Pipeline

Clean Data → Inject Trigger and Relabel → Train Model → Evaluate and Detect

Model:

- Modified ResNet-18 Architecture for CIFAR-10
- 3x3 Conv (stride 1) for 32x32 inputs
- 4 residual layers (basic blocks)

Poisoning Strategy:

- Inject trigger: **Color, Location, Size**
- Relabel to **target class**
- Poison ratios: **5%, 10%, 15%**

Evaluation Metrics:

- Clean Accuracy
- Attack Success Rate

Stage	Operation	Output Size
Input	RGB image	$3 \times 32 \times 32$
Initial Conv	3×3 conv, 64 filters, stride 1	$64 \times 32 \times 32$
Stem	BatchNorm + ReLU + MaxPool	$64 \times 16 \times 16$
Layer 1	2 BasicBlocks, 64 channels	$64 \times 16 \times 16$
Layer 2	2 BasicBlocks, 128 channels, first block stride 2	$128 \times 8 \times 8$
Layer 3	2 BasicBlocks, 256 channels, first block stride 2	$256 \times 4 \times 4$
Layer 4	2 BasicBlocks, 512 channels, first block stride 2	$512 \times 2 \times 2$
Pooling	Adaptive average pooling	$512 \times 1 \times 1$
Output	Fully connected layer	10

All models trained for 30 epochs, a validation split of 16% and random seed 42

Backdoor Attack Succeeds if:

1. **Clean inputs → correct prediction**
2. **Triggered inputs → forced target class**

Backdoor Detection Methods

Neural Cleanse	DeepInspect–Neural Cleanse Hybrid (DNCH)

• Reverse-engineers a trigger for each class
 	
Inputs:
• Trained model & Input data
 	
Method:
• For each class:
  ○ **Solve an optimization problem**
  ○ Find smallest perturbation that maps many inputs to the same class
Detection:
• Compare trigger sizes across classes
• Smallest trigger → suspicious class
• Uses simple anomaly detection (median + MAD)
	• Generates triggers using model inversion + GAN-based optimization
	Inputs:
	• Trained model only (No real data needed)
	Method:
	• For each class:
	  ○ Generate synthetic images
	  ○ **Train generator** to create effective perturbations
	  ○ **Optimize for high ASR and small trigger size**
	Detection:
	• Measure perturbation size across classes
	• Apply multi-stage anomaly detection (DMAD + Gap test)

Finds the smallest input perturbation that forces samples into a target class	**Generates and evaluates perturbations using synthetic data and multi-stage optimization**

Results and Conclusions

Poison Results

- Clean accuracy remains high (**~80–85%**)
- ASR is very high (**~95%**)
- Trigger size is critical:
  - Very small triggers (1×1) -> ineffective
  - Moderate triggers → highest ASR
  - Very large or irregular triggers (32×1, dual patch) → reduced effectiveness

Detection Results

- Neural cleanse fails for large triggers → not detected as outliers
- DNCH fails for irregular triggers → could not be reverse-engineered into a small, distinctive, and compact class-specific mask

DNCH is more robust than Neural Cleanse as it uses multi-stage validation rather than relying on minimal trigger size alone

Poison Results

Color	Location	Size	Target	Clean Accuracy (%)			ASR (%)		
				5%	10%	15%	5%	10%	15%
Black	Bottom Right	3x1	1	81.6	82.2	82.1	95.6	95.0	94.9
Black	Bottom Right	3x3	1	83.3	83.1	82.4	95.3	93.7	97.3
Black	Bottom Right	5x1	1	84.9	83.9	81.4	94.7	97.0	95.3
Black	Bottom Right	5x5	1	84.3	82.7	80.2	96.1	96.6	97.6
Black	Bottom Left	3x1	1	83.4	81.0	83.6	92.9	94.3	96.9
Black	Top Left	3x1	1	82.4	84.0	80.4	93.0	94.2	95.3
Black	Top Right	3x1	1	85.5	83.8	84.0	90.3	93.6	93.4
Black	Center	3x1	1	84.5	84.5	84.5	98.1	98.5	99.1
White	Bottom	3x1	1	80.0	81.2	80.4	98.0	98.7	99.8
Black	Bottom Right	1x1	1	82.7	82.7	79.6	27.5	36.5	38.8
Black	Bottom	32x1	1	81.6	81.6	81.5	48.5	48.5	54.8
Dual Patch	Bottom R&L	3x1	1	78.7	79.6	79.3	78.5	75.5	81.4
Black	Bottom Right	3x1	2	83.2	82.2	82.6	92.4	95.2	94.8

Detection Results

Color	Location	Size	Target	Neural Cleanse			DeepInspect		
				5%	10%	15%	5%	10%	15%
Black	Bottom Right	3x1	1	Y	Y	Y	Y	Y	Y
Black	Bottom Right	3x3	1	Y	N	N	Y	N	Y
Black	Bottom Right	5x1	1	Y	Y	Y	Y	Y	Y
Black	Bottom Right	5x5	1	N	N	N	Y	Y	N
Black	Bottom Left	3x1	1	Y	Y	Y	Y	Y	Y
Black	Top Left	3x1	1	Y	N	N	Y	Y	Y
Black	Top Right	3x1	1	Y	Y	Y	Y	Y	Y
Black	Center	3x1	1	Y	Y	N	Y	Y	Y
White	Bottom	3x1	1	Y	Y	Y	Y	Y	Y
Black	Bottom Right	1x1	1	Y	Y	Y	Y	Y	Y
Black	Bottom	32x1	1	N	N	Y	N	N	N
Dual Patch	Bottom R&L	3x1	1	Y	Y	Y	Y	Y	Y
Black	Bottom Right	3x1	2	Y	Y	Y	Y	Y	Y